

UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES (UITS)

Department of Computer Science & Engineering

CSE402: Internet of Things Lab

Skill Mapping and Justification Form

CO–PO Mapping, Complex Engineering Problem (CEP) and Knowledge Profile (KP)

Justification Report

Assignment/Project Title:	"IoT-Based Medicine Reminder Box: Automated Time-Triggered Medication Alert System Using ESP32 and DS3231 RTC for Elderly and Chronic Care Patients"
Course Code:	CSE402 – Internet of Things Lab
Submitted To:	Md. Zafrul Islam, Lecturer — UITS
Submitted By:	Iqbal Mahmud Emon (0432310005101045), Arafat-Al Sani (0432310005101049), Md Shoyaif Rahman (0432310005101050)
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CO–PO Mapping Table

No.	COs	POs	CP/WP	CA/EA	KP/WK
CO1	Design and implement a basic IoT system using ESP32 microcontroller, explaining the fundamental elements of embedded systems and IoT architecture.	1	-	-	2, 3
CO2	By using modern tools, utilize IoT hardware integration principles, sensor-actuator interfacing, I2C communication protocol, and embedded C/C++ programming.	2	-	-	3
CO3	Demonstrate knowledge by building a team-based IoT project (Medicine Reminder Box) using ESP32, DS3231 RTC, I2C LCD, LEDs, and buzzer as a complete real-world healthcare solution.	3	1,3,6,7	-	4,5,6

Justification of Program Outcome (PO) through Course Outcome (COs)

COs No.	POs	Justification
CO1	PO1 (Engineering Knowledge)	The project requires students to apply embedded systems engineering fundamentals including IoT hardware architecture, microcontroller programming, real-time clock interfacing, I2C bus communication, GPIO configuration, and sensor-actuator logic to solve a real-world healthcare challenge. Students utilize theoretical IoT and embedded systems concepts to integrate the ESP32, DS3231 RTC, 16x2 I2C LCD, color-coded LEDs, and an active buzzer into a fully functional Medicine Reminder Box.
CO2	PO2 (Problem Analysis)	Students analyze a practical healthcare problem — medication non-adherence among elderly and chronically ill patients — and identify the constraints of the solution space: absence of internet connectivity, need for continuous timekeeping across power cycles, limited display output (16x2 LCD), and the requirement for simultaneous multi-modal alerts (visual LED, audio buzzer, and text message). The assignment demands identification of trade-offs in hardware selection, pin mapping conflicts, I2C address management, alarm logic design, and power supply considerations.
CO3	PO3 (Design/Development of Solutions)	The project requires students to design and implement a complete IoT-based medicine reminder system capable of operating autonomously without internet dependency. Students must produce a circuit design integrating multiple I2C devices, write firmware in Arduino C++ implementing event-driven alarm logic, test all functional components against defined pass/fail criteria, and package the solution in a physical medicine storage enclosure. The design satisfies both functional requirements (accurate time-based alerts) and non-functional requirements (compactness, low cost, reliability, and real-world deployability).

Justification of Complex Engineering Problem (CEP) and Knowledge Profile (KP)

Attributes	Justification
P1: Depth of Knowledge	Solving this problem requires in-depth knowledge of embedded systems (K3) including ESP32 microcontroller architecture, I2C communication protocol, GPIO interfacing, real-time clock theory, and interrupt-driven or polling-based event handling. It also demands engineering design knowledge (K4) to architect a layered IoT system (input, processing, output, and physical layers) that is reliable, power-efficient, and deployable in healthcare environments. Students apply engineering practice knowledge (K5) by writing firmware in Arduino C++, validating hardware-software integration, and documenting the system through circuit diagrams, component tables, code explanations, and test results.
P2: Range of Conflicting Requirements	Students must balance conflicting requirements such as low-cost hardware vs. feature richness, hardcoded alarm simplicity vs. user configurability, I2C bus address sharing between the LCD and RTC vs. signal integrity, and buzzer alarm duration sufficiency vs. user disruption. Multiple valid solutions exist (e.g., using Wi-Fi notification instead of buzzer, OLED vs. LCD display, interrupt-driven vs. polling-based alarm detection), requiring trade-off analysis and engineering judgment to select the most practical implementation for the target user group.
P3: Depth of Analysis Required	The project demands detailed analysis of I2C pin sharing between the DS3231 RTC and the 16x2 LCD module, GPIO current limits for LED driving (requiring current-limiting resistors of 220Ω – 330Ω), RTC timekeeping accuracy across power cycles, alarm trigger logic for three independent daily events, and firmware state management between normal operation and alarm mode. Students must justify each hardware and software design decision analytically through test results and documented system architecture rather than relying on trial-and-error or pre-built templates.
P4: Familiarity of Issues	The scenario presents a non-trivial, real-world inspired healthcare IoT application. Students encounter unfamiliar challenges such as maintaining accurate time without internet (NTP) synchronization via a hardware RTC, coordinating multiple I2C slave devices on a shared bus, managing multi-modal output synchronization (LED, buzzer, and LCD simultaneously), and packaging electronics within a physical medicine storage enclosure — contexts not commonly found in standard classroom programming exercises.

P5: Extent of Applicable Codes	Students apply IoT and embedded systems engineering standards including I2C protocol specifications (SDA/SCL signal timing), ESP32 GPIO electrical limits, DS3231 RTC I2C register map, Arduino Library API conventions (RTClib, LiquidCrystal_I2C), and WHO medication adherence guidelines that define the problem context. The source code follows structured embedded C++ conventions with modular function design (runAlarm(), allOff()), appropriate preprocessor definitions for hardware constants, and serial debugging practices.
P6: Extent of Stakeholder Involvement and Conflicting Requirements	The Medicine Reminder Box solution must satisfy multiple stakeholders including elderly patients (need for simple, reliable, non-technical interaction), family caregivers and nurses (need for sufficient alarm intensity and duration), healthcare facilities (need for reliability and low maintenance), patients with chronic conditions such as diabetes and hypertension (need for strict schedule adherence), and students/educators (need for a clear, reproducible IoT demonstration). Each stakeholder group has different operational, accessibility, reliability, and maintenance expectations that influenced the design decisions in the project.
P7: Interdependence	The Medicine Reminder Box integrates interdependent hardware and software subsystems: the DS3231 RTC feeds time data to the ESP32 via I2C; the ESP32 simultaneously drives the I2C LCD, three GPIO-connected LEDs, and an active buzzer; the runAlarm() function coordinates all output devices in a synchronized sequence; and the physical medicine box enclosure provides the contextual deployment layer. Future extensions identified in the report — Wi-Fi push notifications, acknowledgment buttons, cloud data logging, and compartment sensors — further demonstrate the system's interdependence with broader IoT ecosystems including mobile platforms, cloud services, and smart healthcare infrastructure.

Justification of Knowledge Profile (KP)

Knowledge Profile	Justification
K1 – Mathematics and Engineering Fundamentals	Students apply Ohm's Law for current-limiting resistor calculations for LEDs, binary logic for GPIO pin states, timing arithmetic for alarm schedule management, and I2C bus protocol mathematics (address bits, data framing, clock frequency) during the hardware design and firmware development phases of the project.
K2 – Conceptual Understanding of Engineering Sciences	The project requires conceptual understanding of embedded systems architecture — how microcontrollers interface with peripheral devices via communication protocols, how real-time clocks maintain timekeeping independent of main processor state, how I2C bus arbitration supports multiple devices on shared lines, and how event-driven firmware design translates real-world time conditions into physical actuator responses.
K7 – Comprehension of Engineering in Society	The project directly addresses a societal healthcare challenge: medication non-adherence, which the WHO identifies as responsible for significant hospital readmissions and medical complications globally. The Medicine Reminder Box demonstrates engineering's role in improving patient safety, supporting independent living for the elderly, reducing caregiver burden, and lowering healthcare costs through low-cost, accessible technology. It raises awareness of the engineer's professional responsibility to design solutions that are safe, inclusive, and socially beneficial, with due regard for economic constraints of target users in Bangladesh and similar developing-country healthcare contexts.